

AYMANE AIT DADS

Intern, Multimodal Sensing for Smart Home Applications | Multimodal Data Fusion · Time-Series Analysis · Machine Learning

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PROFESSIONAL SUMMARY

Data Science engineering student with hands-on experience building machine learning models on 100K+ real-world sensor records, delivering actionable network-quality insights directly used by an optimization team. Achieved **99.8% accuracy and F1 = 0.999** on a 17,500-image aerial classification task, demonstrating mastery of complex visual dataset analysis and interpretation. Proficient in Python, PyTorch, Scikit-learn, Pandas, and Power BI, with applied experience in feature engineering, clustering pipelines, ensemble methods, and data storytelling for cross-functional stakeholders. Available for internship in Asia; holds valid travel documents with no visa restrictions for Singapore/Southeast Asia region.

CORE COMPETENCIES

Multimodal Sensing | Information Fusion | Time-Series Analysis | Machine Learning Models | Visual Dataset Interpretation | Feature Engineering | Experimental Design | Cross-Functional Collaboration

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed a clustering pipeline combining K-Means and Random Forest on **100K+ fiber-optic network records** (upload speed, latency, jitter), mapping customer performance profiles to 5 commercial service tiers for the network optimization team.
- Engineered automated feature extraction and model evaluation workflows that reduced manual analysis time by **~60%**, directly accelerating the team's maintenance decision cycle.
- Interpreted complex time-series and discrete sensor datasets - upload speed, latency, jitter - to classify network quality across **thousands of nodes**, translating raw signals into actionable maintenance recommendations.
- Delivered stakeholder presentations using **Power BI dashboards**, communicating model outputs as executive-ready business insights to cross-functional team members and product owners.

PROJECTS

Aerial Cactus Detection - Endangered Species Classification

EURECOM · 2025 · 99.8% Accuracy

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images**, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 on endangered-species classification.
- Analyzed and interpreted complex visual datasets in discrete format, applying ensemble methods and ablation studies to optimize model selection for high-stakes ecological sensing applications.

PyTorch · DenseNet121 · CNN · Scikit-learn · Kaggle GPU

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80

- Designed an unsupervised Transformer autoencoder on Mel spectrograms with SpecAugment to detect faults in slide-rail machinery, achieving **AUC 0.80** with no labeled anomaly data available.
- Fused time-series acoustic sensor data with spectrogram visual representations, combining and interpreting multimodal signals to enable predictive maintenance decisions for smart industrial applications.

PyTorch · Transformer Autoencoder · Mel Spectrograms · SpecAugment · NumPy

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Data Science & Analytics: Python | Pandas | NumPy | Scikit-learn | Power BI | SQL | Feature Engineering

Machine Learning & Deep Learning: PyTorch | TensorFlow | CNN | DenseNet | Transformer Models | Ensemble Methods | Clustering

Tools & Infrastructure: Git | Linux | Docker (basic) | PostgreSQL | Matplotlib | Seaborn | Kaggle GPU